CSE382 Hotel Booking Machine Learning - Complete Project Report

Executive summary

This project analyzes the City Hotel (H2) booking demand dataset and addresses data understanding, cleaning, supervised ADR prediction, booking cancellation prediction, customer segmentation, and dimensionality reduction. Tasks 1-5 are recorded in separate repository branches; Task 6 is added as a standalone module. Because the branches are not all integrated into `main`, this report synthesizes branch-specific evidence rather than claiming that one combined end-to-end executable already exists.

Dataset and study scope

The H2 data contains 79,330 rows and 31 columns (Task 1 findings). `ADR` is the regression target and `IsCanceled` the classification target. Approximately 41.73% of bookings were canceled. The study focuses on predictors available at booking creation to avoid information leakage from later outcomes.

Task 1 - Exploratory Data Analysis and Data Cleaning

Branch: `feature/eda`. The team measured four missing Children values, 24 missing Country values, 8,131 literal NULL Agent values, and 75,641 literal NULL Company values. They identified 25,902 duplicate records and 167 zero-guest records. Outlier counts using a 1.5-IQR rule included 3,387 ADR and 2,576 LeadTime values. EDA charts illustrated right-skewed ADR and lead times, cancellation distribution, and their relationships. Cleaning strips whitespace, fills Children with zero, normalizes selected NULLs, and optionally removes zero-guest records; original rows and outliers are otherwise retained.

Task 2 - Preprocessing and Feature Engineering

Branch: `feature/preprocessing`. The team built booking-time feature lists for classification and regression, excluded post-booking information, and engineered TotalNights and TotalGuests. Numerical columns receive median imputation and standardization; categorical columns receive constant imputation and one-hot encoding. GroupShuffleSplit divides data into approximately 70% train, 15% validation, and 15% test while keeping hashed duplicate rows grouped. Preprocessing must be fit on training data only.

Task 3 - Regression and Optimization

Branch: `Regression`. The regression task predicts ADR. The implementation compares Linear Regression, Decision Tree Regression, and Random Forest Regression. Tree and forest variants undergo GridSearchCV; metrics include MAE, RMSE, and R2. Models and the preprocessing object are serialized with joblib. The inspected branch contains source code, but no committed measured regression score table; numeric performance claims are therefore not made for Task 3.

Task 4 - Classification and Optimization

Branch: `feature/classification`. The task predicts booking cancellation with Logistic Regression, Random Forest, and RBF SVM. The baseline report records Logistic Regression validation accuracy 0.7724 and ROC-AUC 0.8199; Random Forest 0.7679 and 0.8311; RBF SVM 0.7748 and 0.8146 (SVM used 5,000 training rows). In the tuning report, Random Forest leads the compared tuned methods: validation accuracy 0.7759, ROC-AUC 0.8529, cancellation F1 0.72. Logistic Regression achieved accuracy 0.7684 and ROC-AUC 0.8202; tuned SVM achieved 0.7591 and 0.8200. Tuning uses validation metrics and grouped cross-validation; final test-set evaluation was not reported.

Task 5 - Clustering and Optimization

Branch: `clustering`. The baseline uses engineered numeric booking attributes and categories, with log-transformation of selected skewed features, scaling, and one-hot encoding. K-Means uses k=4 on 79,163 cleaned bookings, producing 32 encoded features. Its committed baseline metrics show silhouette 0.16406 and

Davies-Bouldin 1.68161; an environment-specific fit time of 0.75 seconds is recorded. The branch includes a PCA 2D cluster plot and a cluster profile CSV. These values describe the baseline, not a claim of globally optimal k.

Task 6 - Dimensionality Reduction and ML Experiments

Branch: `feature/dimensionality-reduction` (prepared handoff). The new script tests PCA retaining 95% variance and LDA (classification only), using train-fitted pipelines. For cancellation prediction, the original 58-dimensional representation achieved validation ROC-AUC 0.79891; PCA reduced 58 to 21 dimensions with ROC-AUC 0.78855; LDA reduced to one dimension with ROC-AUC 0.78885. For ADR, the original 213-dimensional representation achieved RMSE 23.19222 and R2 0.65124; PCA reduced to 28 dimensions with RMSE 28.19475 and R2 0.48456. See the dedicated Task 6 report for full protocol and all metrics.

Integration, caveats, and next steps

The branches use related preprocessing ideas but are not yet a fully merged codebase. The regression branch evaluates baseline and tuned models on the test set during its model loop; future comparisons should instead reserve test data until final model selection. Tasks 4 and 6 use validation reporting and keep the test set unexamined. Grouped random splits mitigate identical-row leakage but do not establish temporal generalization. Future integration should unify the feature policy, record environment versions, use one shared split artifact, and perform a single held-out final test comparison.

Overall conclusion

The team has built complementary analytical components: EDA, structured preprocessing, supervised models, clustering, and dimensionality reduction. Tuned Random Forest is currently strongest among the documented Task 4 classifiers. Task 6 demonstrates that aggressive compression trades away validation accuracy for compact representations in these baseline models. These findings motivate further tuning, integration, and rigorous final testing rather than a universal recommendation for or against reduction.

Evidence sources

Repository: https://github.com/AhmedT-7/CSE382-Hotel-Booking-ML . Branch artifacts: `feature/eda:reports/eda_findi`
`feature/classification:reports/classification_baseline.md`, `feature/classification:reports/classification_t`
`clustering:outputs/clustering/kmeans_baseline_metrics.json`, and corresponding `src/` modules; Task 6 generated outputs and source in the handoff package.